Problem Pack: Exponent Laws and Perfect Squares from MATHEMATICS 8 - UNIT 2 REVIEW

Question 1a

Problem: 1. Simplify each expression. Your final answer should contain only positive exponents.
a) x^4 x^7

Student Work:

x^{4+7} = x^{11}

Feedback

Question 1b

Problem: b) a^{12} ÷ a^5

Student Work:

a^{12-5} = a^7

Feedback

Question 1c

Problem: c) $(m^{3})4$

Student Work:

$(m^{3})4$ = m^{3 } = m^{12}

Feedback

Question 1d

Problem: d) $(2pq^{2})3$

Student Work:

2^3 p^3 q^6 = 8p^3 q^6

Feedback

Question 1e

Problem: e) (3x^2 / 2y)^2

Student Work:

$(3x^{2}/^{2y})2$ = 3^2 $x^{4}/^{4}2$ y^2 = $9x^{4}/^{4y}2$

Feedback

Question 1f

Problem: f) 7r^0

Student Work:
7r^0 = 7(1) = 7

Feedback

Question 2a

Problem: 2. Simplify completely. Use any exponent laws you need, and leave only positive exponents. a) [(2m^3n2)^2(4mn3)] / (8m^2n)

Student Work:
(2m^3 n^2) = 4m^6 n^4 (4m^6 n^4) 16m^2 n^6m^8m2 n = 2m^5 n

Feedback
In question 2a, you squared the first set of parentheses incorrectly by forgetting to square the number 2 inside it. Remember that the exponent outside applies to every single piece on the inside, including the coefficient. Try squaring that 2 first before multiplying everything else out.

Question 2b

Problem: b) (3x^2y)3 / (9x^4y2)

Student Work:
(3x^2y)3 = 27x^6 y^3 27x^6 y^3y^9x4 y^2 = 3x^2 y

Feedback

Question 3

Problem: 3. Classify each number as a perfect square or non-perfect square. If it is a perfect square, give its principal square root. 225 30 4/9 49/20 0.16 0.6

Student Work:
√225 = 15 5^2 < 30 < 6^2 √4/9 = 2/3 √49/20 /20 √0.16 = 0.4 √0.6

Feedback
In question 3, your last fraction part got a bit mixed up because you only applied the square root to the top number instead of the whole fraction. Remember that when a fraction is under a square root, both the top and bottom numbers need to be rooted. Try looking at the bottom number again to see what multiplied by itself makes that denominator.

Question 4a

Problem: 4. State how many real square roots each number has. Briefly explain. a) 81

Student Work:
2 : 9 & −9 = 9^2 = (-9)^2 = 81

Feedback

Question 4b

Problem: b) 0

Student Work:

1 , 0^2 = 0

Feedback

Question 4c

Problem: c) −49

Student Work:

0

Feedback

Question 5

Problem: 5. Evaluate √49. Then list all solutions of x^2 = 49. Explain why the answers are different.

Student Work:

√49 = 7 x^2 = 49 =

Feedback